

9-8

References : See Axler chapter 3 and Strong
Exercise Sections 1.3, 2.4, 2.5, 3.2, 3.3

- 1) Find bases for the kernel and range of cross product operator $T_{\vec{v}}$ defined by

$$T_{\vec{v}}(\vec{x}) = \vec{x} \times \vec{v}$$

where $\vec{v} \neq \vec{0}$.

By a property of the cross product,
 $\ker T_{\vec{v}} = \text{Span}(\vec{v})$. Now as $\langle T_{\vec{v}}(\vec{x}), \vec{v} \rangle = 0$
for all $\vec{x} \in \mathbb{R}^3$, $\text{range } T_{\vec{v}}$ is the plane through
the origin and orthogonal to $\vec{v}$. A basis
for this plane is $\{ \vec{x} \times \vec{v}, (\vec{x} \times \vec{v}) \times \vec{v} \}$
for any $\vec{x} \notin \ker T_{\vec{v}}$.

Counterexample to Theorem 3.8 when V and W
are infinite dimensional:

Let $\mathbb{R}^{\infty} = \{ (x_1, x_2, \dots) : x_i \in \mathbb{R} \text{ for all } i \in \mathbb{N} \}$

and let $R, L \in \mathcal{L}(\mathbb{R}^{\infty}, \mathbb{R}^{\infty})$ be s.t.

$$R(x_1, x_2, \dots) = (0, x_1, x_2, \dots) \quad \text{right shift}$$

$$L(x_1, x_2, \dots) = (x_2, x_3, \dots) \quad \text{left shift}$$

$\ker R = \emptyset$ but R is not surjective

$\ker L = \text{Span}(1, 0, 0, \dots)$ but L is surjective

Matrices (Axler 3.C, Strang 1.3)

Def 4.1 $F^{m \times n}$ is the space of $m \times n$ matrices over F .


Notation: let $A \in F^{m \times n}$ then

$$A = \begin{bmatrix} A_{1,1} & \dots & A_{1,n} \\ \vdots & & \vdots \\ A_{m,1} & \dots & A_{m,n} \end{bmatrix} = [A_{i,j}]_{\substack{j=1,\dots,n \\ i=1,\dots,m}}$$

where each $A_{i,j} \in F$. We will denote the i^{th} row of A as $A_{i,\cdot}$ and the j^{th} column $A_{\cdot,j}$

$$A_{i,\cdot} = [A_{i,1} \quad \dots \quad A_{i,n}]$$

$$A_{\cdot,j} = \begin{bmatrix} A_{1,j} \\ \vdots \\ A_{m,j} \end{bmatrix}$$

Formulas:

$$4.2.i : A_{i,j} = [\vec{e}_i]^T [A\vec{e}_j] = \langle \vec{e}_i, A\vec{e}_j \rangle$$

$$4.2.ii : A_{\cdot,j} = A\vec{e}_j$$

$$4.2.iii : A_{i,\cdot} = (A\vec{e}_i)^T = \vec{e}_i^T A^T$$

Theorem 4.3 $F^{m \times n}$ is an $m \cdot n$ -dimensional vector space over F . The standard basis

$$E_{i,j} = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix}_j^i$$

Theorem 4.4 If V and W are n and m dimensional vector spaces over F , then $\mathcal{L}(V, W) \cong F^{m \times n}$. In other words, given

some basis $\mathcal{B}$ of V and $T \in \mathcal{L}(V, W)$

there exists unique $A \in F^{m \times n}$ s.t. $T(\vec{x}) = A \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$

for any $\vec{x} = \sum_{i=1}^n x_i \vec{v}_i \in V$ where $\vec{v}_i \in \mathcal{B}$.

(Matrices $\longleftrightarrow$ Linear maps)

Matrix Multiplication (Strang 2.4, Axler 3.C)

let $A \in \mathbb{R}^{m \times n}$ and $\vec{x} \in \mathbb{R}^n$ ($\vec{x} \in \mathbb{R}^{n \times 1}$), then $A\vec{x} \in \mathbb{R}^m$ is given by

By rows: $A\vec{x} = \begin{bmatrix} (A_{1,\cdot}) \vec{x} \\ \vdots \\ (A_{m,\cdot}) \vec{x} \end{bmatrix}$

By columns:

$$A\vec{x} = x_1 A_{\cdot,1} + \dots + x_n A_{\cdot,n}$$

let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times l}$ so that $AB \in \mathbb{R}^{m \times l}$. Now let $T_A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ and $T_B \in \mathcal{L}(\mathbb{R}^l, \mathbb{R}^n)$ be such that $T_A \vec{x} = A \cdot \vec{x}$ and $T_B \vec{x} = B \cdot \vec{x}$. Then $AB \cdot \vec{x} = (T_A \circ T_B)(\vec{x})$.

(Axler 3.B, Strang 3.2)

Def 4.4 let $A \in \mathbb{R}^{m \times n}$. The null space and (column space) range of A are defined as

$$\text{null}(A) = \{ \vec{x} \in \mathbb{R}^n : A\vec{x} = \vec{0} \}$$

$$\text{range}(A) = \{ \vec{y} \in \mathbb{R}^m : \vec{y} = A\vec{x} \text{ for some } \vec{x} \in \mathbb{R}^n \}$$

and are subspaces of $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively.

Recall that range A is the span of the column vectors of A and a basis is the set of lin. ind. columns of A .

Exercise:

2) Find bases for $\text{range}(A)$ and $\text{null}(A)$ where

a. $A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & 4 \\ 3 & -1 & 7 \end{bmatrix}$

b. $A = \begin{bmatrix} 2 & 1 & -1 & 0 \\ -4 & 3 & 2 & 1 \end{bmatrix}$

Solutions

a. $\begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & 4 \\ 3 & -1 & 7 \end{bmatrix} \xrightarrow[-3r_1]{-2r_1} \begin{bmatrix} 1 & 0 & 3 \\ 0 & -1 & -2 \\ 0 & -1 & -2 \end{bmatrix}$

$\rightarrow \begin{bmatrix} \textcircled{1} & 0 & 3 \\ 0 & \textcircled{-1} & -2 \\ 0 & 0 & 0 \end{bmatrix}$

range A : look at the pivots

$$\text{range } A = \text{Span} \left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ -1 \end{bmatrix} \right)$$

null A : Set $A\vec{x} = \vec{0}$

$$\begin{aligned} x_1 + 3x_3 &= 0 \\ -x_2 - 2x_3 &= 0 \end{aligned} \Rightarrow \begin{aligned} x_1 &= -3x_3 \\ x_2 &= -2x_3 \end{aligned}$$

$$x_3 = x_3$$

$$\Rightarrow \vec{x} = \begin{bmatrix} -3x_3 \\ -2x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} -3 \\ -2 \\ 1 \end{bmatrix}$$

$$\therefore \text{null } A = \text{span} \left(\begin{bmatrix} -3 \\ -2 \\ 1 \end{bmatrix} \right)$$

$$b. \begin{bmatrix} 2 & 1 & -1 & 0 \\ -4 & 3 & 2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1/2 & -1/10 \\ 0 & 1 & 0 & 1/5 \end{bmatrix}$$

$$\text{range } A = \text{span} \left(\begin{bmatrix} 2 \\ -4 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right) = \mathbb{R}^2$$

$$\text{null } A = \text{span} \left(\begin{bmatrix} 1/2 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1/10 \\ -1/5 \\ 0 \\ 1 \end{bmatrix} \right)$$

Def 4.5 $\text{rank}(A) = \dim(\text{range } A)$ and
 $\text{nullity}(A) = \dim(\text{null } A)$

Theorem 4.6 (Rank-Nullity Theorem) If $A \in \mathbb{R}^{m \times n}$
 then $\text{rank}(A) + \text{nullity}(A) = n$.

Let $A \in \mathbb{R}^{m \times n}$ and assume that $\text{nullity}(A) = k \leq n$.
 Now letting $B_N = \{ \vec{v}_1, \dots, \vec{v}_k \}$ be a basis for
 $\text{null}(A)$, we extend to a basis of $\mathbb{R}^n$ by

taking $\mathcal{B} = \mathcal{B}_N \cup \{\vec{w}_1, \dots, \vec{w}_{n-k}\}$. Now denote $\mathcal{B}_R = \{A\vec{w}_1, \dots, A\vec{w}_{n-k}\}$. We claim that this is a basis for range A . Clearly $\mathcal{B}_R$ span range A . Now assume that

$$\sum \lambda_i A\vec{w}_i = \vec{0}. \quad \text{Then } \sum \lambda_i \vec{w}_i \in \text{null } A \text{ and } A(\sum \lambda_i \vec{w}_i)$$

thus $\sum \lambda_i \vec{w}_i = \vec{0}$ as $\mathcal{B}$ is lin. ind. Finally as $\{\vec{w}_1, \dots, \vec{w}_{n-k}\}$ are lin. ind. as well, $\lambda_i = 0$ for each i and hence $\text{rank } A = n-k$ as desired.

Inverses (Axler 3.D, Strang 2.5)

Def 4.7 $A \in F^{n \times n}$ is invertible iff there exists unique $A^{-1} \in F^{n \times n}$ s.t. $AA^{-1} = A^{-1}A = I_n$ ($n \times n$ identity matrix)

$T \in \mathcal{L}(F^n, F^n)$ is invertible iff there exists unique $T^{-1} \in \mathcal{L}(F^n, F^n)$ s.t. $T \circ T^{-1} = T^{-1} \circ T = \text{id}_{F^n}$

Theorem 4.8 $A \in F^{n \times n}$ is invertible iff $\text{null } A = \emptyset$

Theorem 4.9 If $A, B \in F^{n \times n}$ are invertible then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$

Exercise

- 3) show that the linear map $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^2)$ defined by $T(x_1, x_2) = (2x_1 - x_2, 3x_1 + 4x_2)$ is invertible and find its inverse.

Setting $T(\vec{x}) = \vec{0}$ we get

$$2x_1 - x_2 = 0$$

$$3x_1 + 4x_2 = 0$$

and thus $x_1 = x_2 = 0$. Now as $\ker T = \emptyset$, T is invertible by Theorem 4.8. Now let T^{-1} be s.t. $T^{-1} \circ T = \text{id}$ and set $T(\vec{x}) = \vec{y}$ so that

$$4y_1 = 2x_1 - x_2$$

$$+ y_2 = 3x_1 + 4x_2$$

Then $x_1 = \frac{1}{11}(4y_1 + y_2)$ and

$$x_2 = \frac{1}{11} (-3y_1 + 2y_2). \quad \text{Now as } T^{-1}(\vec{y}) = \vec{x}$$

$$T^{-1}(y_1, y_2) = \frac{1}{11} (4y_1 + y_2, -3y_1 + 2y_2)$$

$$\text{Noting that } T(\vec{x}) = \begin{bmatrix} 2x_1 - x_2 \\ 3x_1 + 4x_2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix} \vec{x}$$

$$\begin{aligned} \text{we see that } T^{-1}(\vec{x}) &= \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}^{-1} \vec{x} \\ &= \frac{1}{11} \begin{bmatrix} 4 & 1 \\ -3 & 2 \end{bmatrix} \end{aligned}$$

Solutions to $A\vec{x} = \vec{b}$ (Strang 3.3)

Theorem 4.10 Let $A \in \mathbb{R}^{n \times n}$ be invertible.

Then $\vec{x} = A^{-1}\vec{b}$ is the unique solution to $A\vec{x} = \vec{b}$ for any $\vec{b} \in \mathbb{R}^n$.

$$A: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

Theorem 4.11 Let $A \in \mathbb{R}^{m \times n}$ be such that $m > n$ and $\text{rank } A = n$. Then for any $\vec{b} \in \text{range } A$ there is a unique solution to $A\vec{x} = \vec{b}$.

Proof: By the Rank-Nullity theorem, $\text{null } A = \emptyset$.

Now let $\vec{b} \in \text{range } A$ and let $\vec{x}_1, \vec{x}_2 \in \mathbb{R}^n$ be s.t. $A\vec{x}_1 = A\vec{x}_2 = \vec{b}$. Then

$$\vec{0} = A\vec{x}_1 - A\vec{x}_2 = A(\vec{x}_1 - \vec{x}_2)$$

and thus $\vec{x}_1 - \vec{x}_2 \in \text{null } A$. Hence $\vec{x}_1 - \vec{x}_2 = \vec{0}$ and $\vec{x}_1 = \vec{x}_2$ as desired.